Analytic Signal Representation via Explicit Harmonic Conjugation of a Meromorphic Generator

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Abstract

We introduce a class of meromorphic functions for which both the logarithmic potential and its harmonic conjugate admit explicit closed-form expressions. This allows direct construction of the associated analytic signal bypassing the need for explicit Hilbert transform evaluation as a singular integral. Instead, the analytic signal is obtained via explicit complex continuation. The underlying structure is governed by a logarithmic Green potential generated by a periodic lattice of poles.

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1 Introduction

The Hilbert transform plays a central role in harmonic analysis and signal processing, notably in the construction of analytic signals. However, its definition as a singular integral often makes both theoretical analysis and numerical evaluation challenging.

In this work, we exhibit a class of meromorphic functions for which the analytic signal can be obtained without computing any singular integral. This is achieved by exploiting explicit closed-form expressions of the logarithmic potential and its harmonic conjugate in the complex plane.

The analysis is carried out in the setting of analytic functions in strip domains and Hardy space boundary values, and it lies at the interface between complex analysis and harmonic analysis, with applications to analytic signal construction.

This paper should be viewed as a fully explicit model case within a broader framework developed elsewhere, where the present function plays the role of a tractable prototype illustrating the general mechanism.

2 Meromorphic Generator

We consider the function

$$f(z;\alpha) = \frac{e^{\alpha z}}{\cosh z}, \qquad \alpha \in (-1,1), \quad z = x + iy. \tag{1}$$

The poles of f are located at

$$z_n = i\left(\frac{\pi}{2} + n\pi\right), \qquad n \in \mathbb{Z}, \tag{2}$$

and f is analytic in the strip $|\operatorname{Im}(z)| < \pi/2$.

3 Explicit Potential and Harmonic Conjugate

Define the logarithmic potential

$$U(x, y) = \log |f(x + iy)|. \tag{3}$$

3.1 Main Result

Theorem 1. *The functions U and its harmonic conjugate V admit the explicit forms:*

$$U(x, y) = \alpha x - \frac{1}{2} \log \big(\cosh(2x) + \cos(2y) \big) + C, \tag{4}$$

$$V(x, y) = \alpha y - \arctan(\tanh x \tan y) + C', \tag{5}$$

where C and C' are real constants depending on the branch of the logarithm.

Proof.
Fixing the principal branch of the logarithm in the simply connected strip $|\operatorname{Im}(z)| < \pi/2$, which contains no zeros of $\cosh z$, ensuring that F is holomorphic, we write

$$\log f(z) = \alpha z - \log(\cosh z).$$

Using

$$\cosh(x + iy) = \cosh x \cos y + i \sinh x \sin y,$$

one obtains

$$|\cosh(x + iy)|^2 = \frac{1}{2}(\cosh(2x) + \cos(2y)).$$

This yields the expression for U. The argument of $\cosh(x + iy)$ is

$$\arg(\cosh(x + iy)) = \arctan(\tanh x \tan y),$$

which gives V. Consequently, $F(z) = \alpha z - \log(\cosh z)$ is holomorphic in the strip and therefore satisfies the Cauchy–Riemann equations. □

4 Analytic Signal Representation

Let $F(z) = \log f(z) = U + iV$. Since F is analytic in the strip, its boundary values satisfy the standard harmonic conjugation relations, which follows from the boundary value theory of Hardy spaces in strip domains and the Plemelj–Sokhotski formulas.

Theorem 2. *For $x \in \mathbb{R}$,*

$$f(x) + i\,\mathcal{H}[f](x) = \lim_{\epsilon \to 0^+} \exp \big(U(x, \epsilon) + iV(x, \epsilon) \big), \tag{6}$$

where $\mathcal{H}$ denotes the Hilbert transform, defined in the distributional sense after subtraction of the linear term αx.

Proof. Since $F(z)$ is analytic in any simply connected subdomain of the upper half-strip avoiding the zeros of cosh z, with a consistent branch of the logarithm, the boundary values satisfy

$$V(x, 0) = \mathcal{H}[U(\cdot, 0)](x)$$

under standard regularization after subtraction of the linear term αx to ensure admissibility in the distributional sense. The result follows from $f(z) = \exp(F(z))$, taking the non-tangential limit and follows from standard boundary value theory of analytic functions (see e.g. classical results on Hardy spaces). $\square$

Remark 1 (Regularity). The function $U(x, 0)$ exhibits linear growth due to the term αx and therefore does not belong to $L^2(\mathbb{R})$. The Hilbert transform is understood in a distributional sense after subtraction of this linear term αx to ensure admissibility in the distributional sense.

5 Harmonic Structure

The potential U satisfies

$$\Delta U = -2\pi \sum_{n \in \mathbb{Z}} \delta(z - z_n) \tag{7}$$

in the sense of distributions. Thus, U is a logarithmic Green potential generated by a periodic lattice of poles. This follows from standard distributional properties of logarithmic potentials associated with meromorphic functions.

6 Discussion

The results provide an explicit instance where the analytic signal can be constructed without computing the Hilbert transform as a singular integral. Instead, the transform is realized through analytic continuation in the complex plane.

This framework highlights a direct connection between:

- meromorphic structure,
- harmonic conjugation,
- analytic signal construction.

Possible extensions include other meromorphic generators with structured pole distributions.

7 Conclusion

We have exhibited a class of meromorphic functions for which the logarithmic potential and its harmonic conjugate admit closed-form expressions. This enables a direct reconstruction of the analytic signal via complex continuation, bypassing the need for explicit Hilbert transform evaluation.

References

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